

0.1 Construction of Hamiltonian

rename this section to something like model?

0.1.1 Construction of the Hamiltonian

The Hamiltonian is

$$\mathcal{H} = \sum_{\langle ij \rangle} J_{ij} \left(\vec{S}_i \cdot \vec{S}_j + \kappa \left(\vec{S}_i \cdot \vec{S}_j \right)^2 \right) \quad (1)$$

Using the Abrikosov representation of the spin in terms of fermions:

$$\vec{S}_i = \sum_{\alpha\beta} f_{i\alpha}^\dagger \frac{\vec{\sigma}}{2} f_{i\beta} \quad (2)$$

$$\begin{aligned} \Rightarrow \vec{S}_i \cdot \vec{S}_j &= \frac{1}{4} \sum_{\alpha\beta\gamma\delta} f_{i\alpha}^\dagger f_{i\beta} f_{j\gamma}^\dagger f_{j\delta} \underbrace{\vec{\sigma}_{\alpha\beta} \vec{\sigma}_{\gamma\delta}}_{=\delta_{\alpha\delta}\delta_{\beta\gamma} - \delta_{\alpha\gamma}\delta_{\beta\delta}} \\ &= \frac{1}{4} \sum_{\alpha\beta} 2f_{i\alpha}^\dagger f_{i\beta} f_{j\beta}^\dagger f_{j\alpha} - f_{i\alpha}^\dagger f_{i\alpha} f_{j\beta}^\dagger f_{j\beta} \end{aligned}$$

Move the last annihilation operator in the first term to the second position: use $f_{j\beta}^\dagger f_{j\alpha} = \delta_{\alpha\beta} - f_{j\alpha} f_{j\beta}^\dagger$ and acquire 1 minus signs. After that bring the other annihilation operator to the last position which gives another minus.

$$\vec{S}_i \cdot \vec{S}_j = \frac{1}{4} \sum_{\alpha\beta} -2f_{i\alpha}^\dagger f_{j\alpha} f_{j\beta}^\dagger f_{j\beta} + 2f_{i\alpha} f_{i\beta} \delta_{\alpha\beta} - f_{i\alpha}^\dagger f_{i\alpha} f_{j\beta}^\dagger f_{j\beta} \quad (3)$$

The second term is n_i and the third is $n_i n_j$. For the first term I introduce the shorthand

$$\chi_{ij} = \frac{1}{2} \sum_{\alpha} f_{i\alpha}^\dagger f_{j\alpha} \quad (4)$$

$$\vec{S}_i \cdot \vec{S}_j = -2\chi_{ij}\chi_{ji} + \frac{1}{2}n_i - \frac{1}{4}n_i n_j \quad (5)$$

Using the constraint $n_i = n_j = 1$, the final expression is

$$\vec{S}_i \cdot \vec{S}_j = -2\chi_{ij}\chi_{ji} + \frac{1}{4} \quad (6)$$

Relevant for the Hamiltonian is also the term:

$$\Rightarrow \left(\vec{S}_i \cdot \vec{S}_j \right)^2 = \frac{1}{16} - \chi_{ij}\chi_{ji} + 4(\chi_{ij}\chi_{ji})^2 \quad (7)$$

The mean field decoupling works like

$$\chi_{ij}\chi_{ji} \approx \langle \chi_{ij} \rangle \chi_{ji} + \langle \chi_{ji} \rangle \chi_{ij} - |\langle \chi_{ij} \rangle|^2 \quad (8)$$

$$(\chi_{ij}\chi_{ji})^2 \approx 2|\langle \chi_{ij} \rangle|^2 \langle \chi_{ij} \rangle \chi_{ji} + 2|\langle \chi_{ij} \rangle|^2 \langle \chi_{ji} \rangle \chi_{ij} - 3|\langle \chi_{ij} \rangle|^4 \quad (9)$$

$$\Rightarrow \vec{S}_i \cdot \vec{S}_j \approx -2\langle \chi_{ij} \rangle \chi_{ji} + \text{h.c.} + 2|\langle \chi_{ij} \rangle|^2 + \frac{1}{4} \quad (10)$$

$$\Rightarrow \left(\vec{S}_i \cdot \vec{S}_j \right)^2 \approx -\langle \chi_{ij} \rangle \chi_{ji} + 8|\langle \chi_{ij} \rangle|^2 \langle \chi_{ij} \rangle \chi_{ji} + \text{h.c.} + |\langle \chi_{ij} \rangle|^2 - 12|\langle \chi_{ij} \rangle|^4 + \frac{1}{16} \quad (11)$$

Maybe comment of the fact that the mean-field parameters know about the fermi sea and thereby about many states. In this sense the states do speak with each other. Bringing everything together:

$$\begin{aligned} \mathcal{H} \approx \sum_{\langle ij \rangle} J_{ij} & \left[\left(-1 - \kappa/2 + 4\kappa |\langle \chi_{ij} \rangle|^2 \right) \langle \chi_{ij} \rangle \left(f_{j\uparrow}^\dagger f_{i\uparrow} + f_{j\downarrow}^\dagger f_{i\downarrow} \right) + \text{h.c} \right. \\ & \left. + \left(2 + \kappa - 12\kappa |\langle \chi_{ij} \rangle|^2 \right) |\langle \chi_{ij} \rangle|^2 + \frac{1}{4} + \frac{\kappa}{16} \right] \end{aligned} \quad (12)$$

It is sometimes handy to define

$$\tilde{J}_{ij} = J_{ij} (1 + \kappa/2), \quad \tilde{\kappa} = \kappa/(1 + \kappa/2) \quad (13)$$

I drop the constant:

$$\begin{aligned} \mathcal{H} \approx \sum_{\langle ij \rangle} \tilde{J}_{ij} & \left[- \left(1 - 4\tilde{\kappa} |\langle \chi_{ij} \rangle|^2 \right) \langle \chi_{ij} \rangle \left(f_{j\uparrow}^\dagger f_{i\uparrow} + f_{j\downarrow}^\dagger f_{i\downarrow} \right) + \text{h.c} \right. \\ & \left. + 2 \left(1 - 6\tilde{\kappa} |\langle \chi_{ij} \rangle|^2 \right) |\langle \chi_{ij} \rangle|^2 \right] \end{aligned} \quad (14)$$

This complies with the result obtained here [2]. Hence, the mean field produces the effective hopping amplitude t^{MF} :

$$t_{ji}^{\text{MF}} = -\tilde{J}_{ij} \langle \chi_{ij} \rangle \left(1 - 4\tilde{\kappa} |\langle \chi_{ij} \rangle|^2 \right) = -J_{ij} \langle \chi_{ij} \rangle \left(1 + \kappa/2 - 4\kappa |\langle \chi_{ij} \rangle|^2 \right). \quad (15)$$

$$\Rightarrow \mathcal{H} \approx \sum_{\langle ij \rangle} t_{ji}^{\text{MF}} \left(f_{j\uparrow}^\dagger f_{i\uparrow} + f_{j\downarrow}^\dagger f_{i\downarrow} \right) + \text{const.} \quad (16)$$

This is pretty interesting: the hopping-amplitude $i \rightarrow j$ is directly proportional to the hopping-amplitude $j \rightarrow i$. **Here are a few ideas how to make use/discuss this insight:**

- Maybe one can also make the argument that we start with a "symmetric" model, i.e. a model without internal sense of direction, a feature that is now preserved in the Mean field Hamiltonian.
- Or maybe, even more sophisticated, one could make argument using structure of the spin-singlet states.
- Maybe one can use this fact to simplify the Mean-field equation?

0.1.2 Continuity equation and probability current

The probability (/particle) density on site k is

$$n_k = \sum_{\sigma} f_{k\sigma}^\dagger f_{k\sigma}. \quad (17)$$

Now I derive the corresponding probability current density:

$$\frac{\partial n_k}{\partial t} = -i \sum_{\sigma} \left([f_{k\sigma}^\dagger, H] f_{k\sigma} + f_{k\sigma}^\dagger [f_{k\sigma}, H] \right) \quad (18)$$

$$\begin{aligned} [f_{k\sigma}^\dagger, H] f_{k\sigma} &= - \sum_{\delta} t_{k+\delta; k} f_{k+\delta; \sigma}^\dagger f_{k\sigma} - \mu_k f_{k\sigma}^\dagger f_{k\sigma} \\ f_{k\sigma}^\dagger [f_{k\sigma}, H] &= \sum_{\delta} t_{k; k+\delta} f_{k\sigma}^\dagger f_{k+\delta; \sigma} + \mu_k f_{k\sigma}^\dagger f_{k\sigma} \end{aligned} \quad (19)$$

$$\Rightarrow \frac{\partial n_k}{\partial t} = - \sum_{\delta} i \left(\sum_{\sigma} t_{k,k+\delta} f_{k\sigma}^{\dagger} f_{k+\delta;\sigma} - t_{k+\delta,k} f_{k+\delta;\sigma}^{\dagger} f_{k\sigma} \right) \quad (20)$$

Define

$$J_{k \rightarrow k+\delta} \equiv i \left(\sum_{\sigma} t_{k,k+\delta} f_{k\sigma}^{\dagger} f_{k+\delta;\sigma} - t_{k+\delta,k} f_{k+\delta;\sigma}^{\dagger} f_{k\sigma} \right) \quad (21)$$

$J_{k \rightarrow k+\delta}$ is the (directed!) current density along the bond $\vec{R}_k \rightarrow \vec{R}_{k+\delta}$. The discrete (/lattice) continuity equation reads

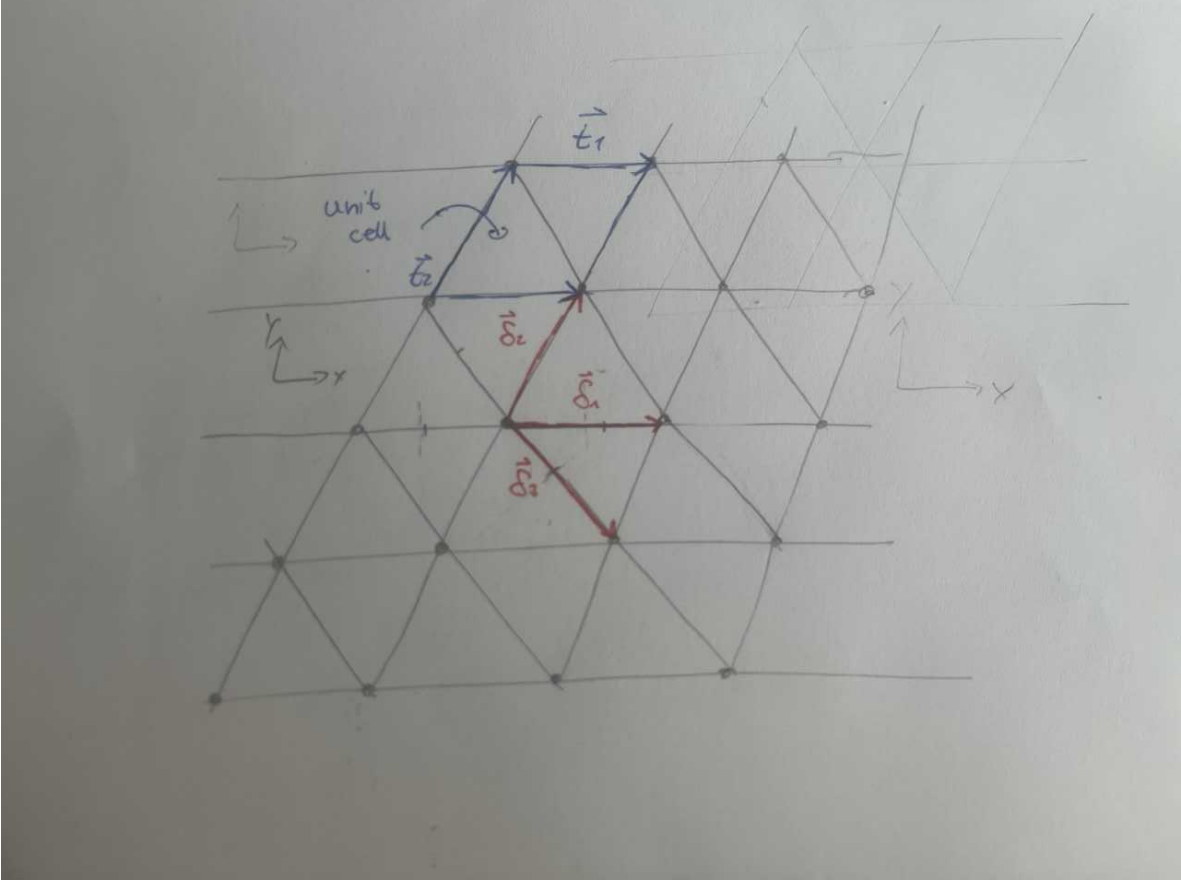
$$\frac{\partial n_k}{\partial t} = - \sum_{\delta} J_{k \rightarrow k+\delta}. \quad (22)$$

Something interesting happens when calculating $\langle J_{k \rightarrow k+\delta} \rangle$:

$$\begin{aligned} \langle J_{k \rightarrow k+\delta} \rangle &= i \left(t_{k,k+\delta} \left\langle \sum_{\sigma} f_{k\sigma}^{\dagger} f_{k+\delta;\sigma} \right\rangle - t_{k+\delta,k} \left\langle \sum_{\sigma} f_{k+\delta;\sigma}^{\dagger} f_{k\sigma} \right\rangle \right) \\ &= 2i (t_{k,k+\delta} \langle \chi_{k,k+\delta} \rangle - t_{k+\delta,k} \langle \chi_{k+\delta,k} \rangle) \\ &= -2i \tilde{J}_{k;k+\delta} \left(1 - 4\tilde{\kappa} |\langle \chi_{k,k+\delta} \rangle|^2 \right) \left(\langle \chi_{k+\delta;k} \rangle \langle \chi_{k;k+\delta} \rangle - \langle \chi_{k;k+\delta} \rangle \langle \chi_{k+\delta;k} \rangle \right) \\ &= 0. \end{aligned} \quad (23)$$

Apparently, the current on each bond vanishes in the ground state by construction. **This still leaves the possibility that single states have non-zero probability currents. However, it remains to investigate how this is possible for e.g. states with a singular (topologically protected) edge mode. One would guess that this feature would lead to a non-zero current in the ground state.**

0.2 0-0 and π - π States



This system is a simple triangular Bravais lattice with primitive translations

$$\vec{a}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{a}_2 = \begin{pmatrix} 1/2 \\ \sqrt{3}/2 \end{pmatrix}. \quad (24)$$

0.2.1 Pedestrian method

The tight-binding Hamiltonian is:

$$\begin{aligned} H &= \sum_{i,\delta,\sigma} \frac{t}{2} \left(f_{i\sigma}^\dagger f_{i+\delta\sigma} + \text{h.c.} \right) + \mu \sum_i \left(f_{i\uparrow}^\dagger f_{i\uparrow} + f_{i\downarrow}^\dagger f_{i\downarrow} - 1 \right) + \text{const.} \\ &= \sum_{i,\delta,\sigma} \frac{t}{2} \left(f_{i\sigma}^\dagger f_{i+\delta\sigma} + \text{h.c.} \right) + \mu \sum_{i,\sigma} f_{i\sigma}^\dagger f_{i\sigma} - \mu N + \text{const..} \end{aligned} \quad (25)$$

i runs over sites and δ over nearest neighbours of a generic site. I include the absolute shift for the sake of consistency. The factor 1/2 is also included for consistency. The constant is a product of mean field theory and reads

$$\frac{1}{2} \sum_{i,\delta} 2\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} \rangle|^2 \right) |\langle \chi_{i,i+\delta} \rangle|^2 = 6N \left(\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} \rangle|^2 \right) \right) |\langle \chi_{i,i+\delta} \rangle|^2 \quad (26)$$

The second step can be understood in at least two ways; a) each site has 6 neighbours, or b) there are bond per primitive unit cell 1 of which is completely inside the unit cell and contributes a 2 and the

other 4 shared between 2 unit cells thereby contributing a 1. All bonds are equivalent of course. The aforementioned six nearest neighbours lie at the (relative) positions

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\delta}_2 = \begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix} \quad \vec{\delta}_3 = \begin{pmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{pmatrix}$$

and $-\vec{\delta}_1, -\vec{\delta}_2, -\vec{\delta}_3$. I will start with the 0-0 state. To advance, do a Fourier transform:

$$f_{i\sigma} = \frac{1}{\sqrt{N}} \sum_k e^{i\vec{k}\vec{R}_i} c_{k\sigma}. \quad (27)$$

$$\begin{aligned} \Rightarrow H &= \sum_{q,\sigma} \frac{t}{2} \left(c_{q\sigma}^\dagger c_{q\sigma} \left(\sum_{\delta} e^{i\vec{q}\vec{\delta}} + \mu \right) + \text{h.c.} \right) - \mu N \\ &= \sum_{q,\sigma} \left(t \sum_{\delta} \cos(\vec{q}\vec{\delta}) + \mu \right) c_{q\sigma}^\dagger c_{q\sigma} - \mu N \end{aligned} \quad (28)$$

Using first the symmetrie of the cosine and then the sum identity for the cosine the dispersion reads

$$\begin{aligned} \varepsilon(\vec{q}) &= 2t \left[\cos(q_x) + \cos\left(\frac{1}{2}(q_x + \sqrt{3}q_y)\right) + \cos\left(\frac{1}{2}(q_x - \sqrt{3}q_y)\right) \right] + \mu \\ &= 2t \left[\cos(q_x) + 2 \cos\left(\frac{q_x}{2}\right) \cos\left(\frac{\sqrt{3}q_y}{2}\right) \right] + \mu \end{aligned} \quad (29)$$

features:

- $\varepsilon(\vec{q}) \in [-3t, 6t] + \mu \Rightarrow \mu \in [-6t, 3t]$
- symmetries $q_x \rightarrow -q_x$ and $q_y \rightarrow -q_y$ (+ should have some rotational symmetrie and (thereby associated) inflections along other axes) **could check this**
- periodicity: 4π in q_x , $(4\pi)/\sqrt{3}$ in q_y

The mean field parameters are

$$\langle \chi_{ij} \rangle = \frac{1}{2} \langle f_{i\uparrow}^\dagger f_{j\uparrow} + f_{i\downarrow}^\dagger f_{j\downarrow} \rangle = \langle f_{i\uparrow}^\dagger f_{j\uparrow} \rangle \quad (30)$$

Simplify the expression by Fourier transform:

$$\begin{aligned} \langle \chi_{ij} \rangle &= \frac{1}{N} \sum_{q,p} e^{i(\vec{q}\vec{R}_j - \vec{k}\vec{R}_i)} \langle c_p^\dagger c_q \rangle \\ &= \frac{1}{N} \sum_{\vec{q}} e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} f(\varepsilon(\vec{q})) \end{aligned} \quad (31)$$

Take the thermodynamic limit $\sum_k \rightarrow (N\Omega)/(2\pi)^2 \int dk$ (Ω is the unit cell volume).

$$\Rightarrow \langle \chi_{ij} \rangle = \Omega \int_{\text{1.BZ}} \frac{d^2 q}{(2\pi)^2} e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} f(\varepsilon(\vec{q})) \quad (32)$$

It looks like the model allows for different χ_{ij} in each direction $\vec{R}_j - \vec{R}_i$ (there are 3 of those)?

From this expression, recover the on-site particle number by setting $i = j$ and accounting for the factor 1/2 in the definition of χ_{ij} :

$$\langle N_i \rangle = 2\Omega \int \frac{d^2 q}{(2\pi)^2} f(\varepsilon(\vec{q})). \quad (33)$$

By construction the particle number is uniform.

Some elaborations on the integral domain follow. The primitive translations of the reciprocal lattice are

$$\vec{b}_1 = 2\pi \begin{pmatrix} 0 \\ 2/\sqrt{3} \end{pmatrix}, \quad \vec{b}_2 = 2\pi \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix} \quad (34)$$

I do a basis transformation onto those axes defined by

$$\begin{pmatrix} k_x \\ k_y \end{pmatrix} = \begin{pmatrix} 1/2 & \sqrt{3}/2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} q_x \\ q_y \end{pmatrix} \quad (35)$$

$$\Leftrightarrow \begin{pmatrix} q_x \\ q_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 2/\sqrt{3} & -1/\sqrt{3} \end{pmatrix} \begin{pmatrix} k_x \\ k_y \end{pmatrix} \quad (36)$$

$$\Rightarrow \det J = -2/\sqrt{3} \quad (37)$$

$$\Rightarrow \varepsilon(\vec{k}) = 2t (\cos(k_x) + \cos(k_y) + \cos(k_x - k_y)) + \mu \quad (38)$$

The first Brillouin zone is $\{k_x, k_y\} \in [-\pi, \pi]^2$. The unit cell volume is $\Omega = \sqrt{3}/2$. Consistency check: $|\det(J)| = \Omega^{-1}$, since the substitution simply transforms the unit cell to a square of unit volume. So the integrals that have to be evaluated are of the form

$$\langle N_i \rangle = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} f(\varepsilon(\vec{k}), \mu), \quad (39)$$

$$\langle \chi_{ij} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \exp \left[i \left(k_y, (1/\sqrt{3})(2k_x - k_y) \right) \cdot (\vec{R}_j - \vec{R}_i) \right] f(\varepsilon(\vec{k}), \mu) \quad (40)$$

Check that under a particle hole transformation χ changes sign, sign of t is doing just that and this brings the π -flux phase.

The condition is $\langle N_i \rangle \stackrel{!}{=} 1 \forall i$. By setting $t = -1$, the chemical potential is numerically determined in units of t using Mathematica. I choose $t = -1$ because I anticipate t to be negative for a positive χ . (This may not be true depending on κ and χ , but as long as it is self-consistent everything is fine.) For the numerical integration I used a Fermi-function with a small temperature ($\beta = 200t$) instead of a step-function, since the ladder is not analytical and causes problems for the numerical algorithm. The result is $\mu = -0.834737 t$. The bond-parameters are calculated the same way, the result is $\langle \chi_{ij} \rangle = 0.164713$.

Now, it is possible to express μ only in units of J . Use the formula

$$t_{ij} = -J \langle \chi_{ij} \rangle^* (1 + \kappa/2 - 4\kappa |\langle \chi_{ij} \rangle|^2). \quad (41)$$

Set $\kappa = 10$:

$$t_{ij} = -0.8095 J \quad (42)$$

$$\mu = -0.67574 J \quad (43)$$

$$\langle \chi_{ij} \rangle = 0.164713 \quad (44)$$

The energy density (that is also the free energy density at $T = 0$) is

$$\frac{\langle H \rangle}{N} = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \varepsilon(\vec{k}) f(\varepsilon(\vec{k})) - \mu + 6\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i, i+\delta} \rangle|^2 \right) |\langle \chi_{i, i+\delta} \rangle|^2 = -0.888304 J \quad (45)$$

Observation; the chemical potential drops out of the Free energy.

One can also directly calculate results for the state with all fluxes π by performing a particle-hole-transformation:

$$t \rightarrow -t \quad (46)$$

$$\mu \rightarrow -\mu \quad (46)$$

$$\Rightarrow \varepsilon(\vec{k}) \rightarrow -\varepsilon(\vec{k}) \quad (47)$$

$$\Rightarrow f(\varepsilon(\vec{k})) \rightarrow 1 - f(\varepsilon(\vec{k})) \quad (48)$$

It can be shown that

$$\chi \rightarrow -\chi. \quad (49)$$

The result is:

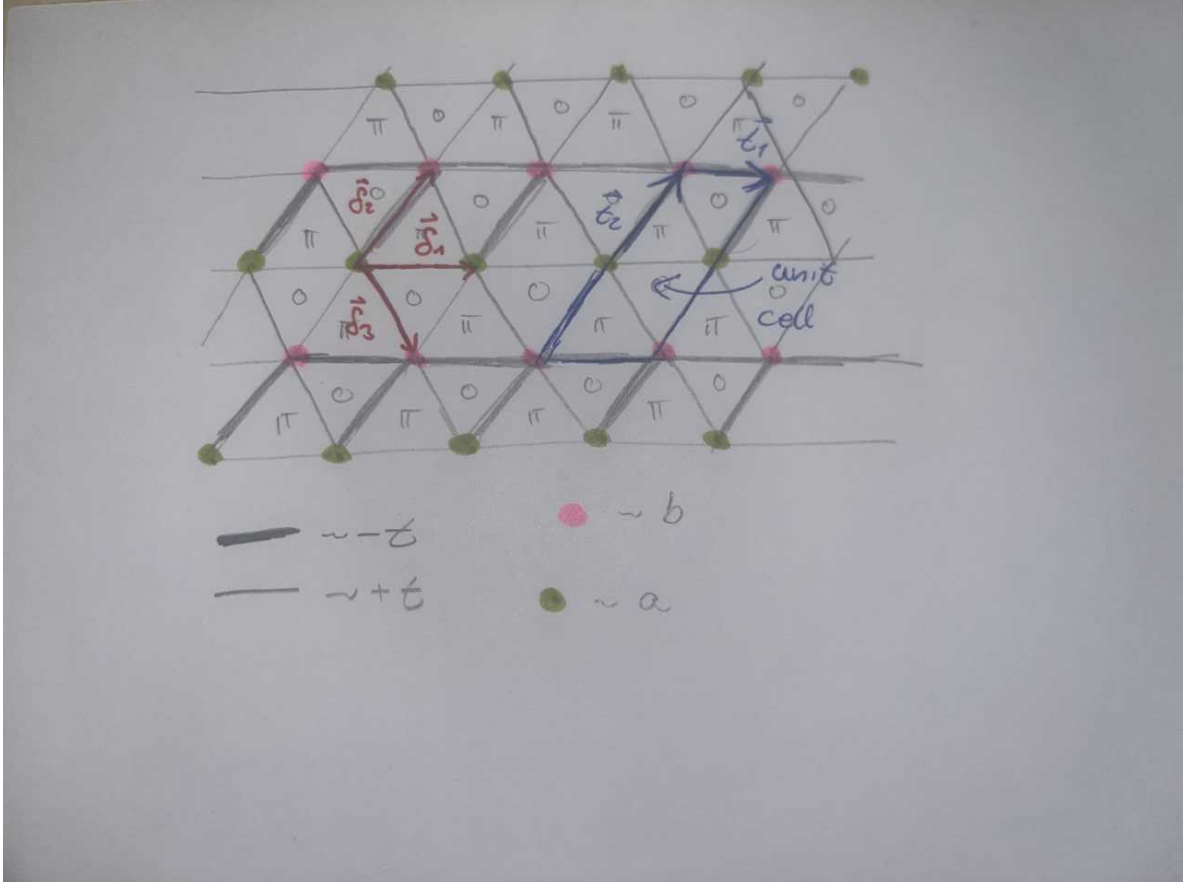
$$\frac{\langle H \rangle_\pi}{N} = -0.888302 J \quad (50)$$

0.2.2 Pedro's method

The longer explanation for this method is in the next section. Here I will only say that the free energy is the same except for γ , which gets replaced by a factor ϕ (as mentioned above the chemical potential does not contribute to the free energy):

0.3 0- π and π -0 States

0.3.1 pedestrian method



The system is a triangular bravais lattice with a two-atomic basis. The primitive translations are

$$\vec{a}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{a}_2 = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad (51)$$

Hence, two types of lattice sites exist, a and b (see sketch). Also χ_{ij} , and thereby t_{ij} can change sign (see sketch). The Hamiltonian is

$$H = \frac{1}{2} \sum_{\langle i,j \rangle, \sigma} \left(t a_{i\sigma}^\dagger a_{j\sigma} - t b_{i\sigma}^\dagger b_{j\sigma} + t_{ij}^{(ab)} a_{i\sigma}^\dagger b_{j\sigma} + \text{h.c.} \right) + \mu_a \sum_{i,\sigma} \left(a_{i\sigma}^\dagger a_{i\sigma} - 1 \right) + \mu_b \sum_{i,\sigma} \left(b_{i\sigma}^\dagger b_{i\sigma} - 1 \right) + \text{const.} \quad (52)$$

$$\begin{aligned} H = \frac{t}{2} \sum_{i\sigma} & \left(a_{i\sigma}^\dagger a_{i+\delta_1;\sigma} + a_{i\sigma}^\dagger a_{i-\delta_1;\sigma} - a_{i\sigma}^\dagger b_{i+\delta_2;\sigma} + a_{i\sigma}^\dagger b_{i-\delta_2;\sigma} + a_{i\sigma}^\dagger b_{i+\delta_3;\sigma} + a_{i\sigma}^\dagger b_{i-\delta_3;\sigma} \right. \\ & \left. - b_{i\sigma}^\dagger b_{i+\delta_1;\sigma} - b_{i\sigma}^\dagger b_{i-\delta_1;\sigma} + b_{i\sigma}^\dagger a_{i+\delta_2;\sigma} - b_{i\sigma}^\dagger a_{i-\delta_2;\sigma} + b_{i\sigma}^\dagger a_{i+\delta_3;\sigma} + b_{i\sigma}^\dagger a_{i-\delta_3;\sigma} \right. \\ & \left. + \mu_a a_{i\sigma}^\dagger a_{i\sigma} + \mu_b b_{i\sigma}^\dagger b_{i\sigma} \right) - \frac{N(\mu_a + \mu_b)}{2} \end{aligned} \quad (53)$$

The site index notation is not ideal, as a and b sites by definition cant have the same site indices. The constant is the same as for the 0-flux case, provided that the absolute value of $\chi_{i,i+\delta}$ is uniform.

$$\frac{1}{2} \sum_{i,\delta} 2\tilde{J} \left(1 - 6|\langle\chi_{i,i+\delta}\rangle|^2\right) |\langle\chi_{i,i+\delta}\rangle|^2 = 6N\tilde{J} \left(1 - 6\tilde{\kappa}|\langle\chi_{i,i+\delta}\rangle|^2\right) |\langle\chi_{i,i+\delta}\rangle|^2 \quad (54)$$

This is not quite right yet. The Fourier transformation has to be modified bc there are only $N/2$ of a and b-sites respectively

$$a_{i\sigma} = \sqrt{\frac{2}{N}} \sum_k e^{i\vec{k}\vec{R}_i} a_{k\sigma}, \quad (55)$$

and the same for $b_{i\sigma}$. The Hamiltonian (without the constants) becomes

$$H = \sum_{q,\sigma} \begin{pmatrix} a_{q\sigma}^\dagger & b_{q\sigma}^\dagger \end{pmatrix} \begin{pmatrix} 2t \cos(\vec{q}\vec{\delta}_1) + \mu_a & 2t \cos(\vec{q}\vec{\delta}_3) - 2it \sin(\vec{q}\vec{\delta}_2) \\ 2t \cos(\vec{q}\vec{\delta}_3) + 2it \sin(\vec{q}\vec{\delta}_2) & -2t \cos(\vec{q}\vec{\delta}_1) + \mu_b \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \end{pmatrix}. \quad (56)$$

The primitive translations of the reciprocal lattice are

$$\vec{b}_1 = 2\pi \begin{pmatrix} 0 \\ 1/\sqrt{3} \end{pmatrix}, \quad \vec{b}_2 = 2\pi \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix} \quad (57)$$

The suitable transformation is

$$\begin{pmatrix} k_x \\ k_y \end{pmatrix} = \begin{pmatrix} 1 & \sqrt{3} \\ m1 & 0 \end{pmatrix} \begin{pmatrix} q_x \\ q_y \end{pmatrix} \quad (58)$$

$$\Leftrightarrow \begin{pmatrix} q_x \\ q_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1/\sqrt{3} & -1/\sqrt{3} \end{pmatrix} \begin{pmatrix} k_x \\ k_y \end{pmatrix} \quad (59)$$

$$\Rightarrow \det(J) = 1/\sqrt{3} \quad (60)$$

This yields

$$\begin{aligned} \cos(\vec{q}\vec{\delta}_1) &\rightarrow \cos(k_y) \\ \sin(\vec{q}\vec{\delta}_2) &\rightarrow \sin(k_x/2) \end{aligned} \quad (61)$$

$$\cos(\vec{q}\vec{\delta}_3) \rightarrow \cos(k_y - k_x/2) \quad (62)$$

$$\varepsilon(\vec{k})_{\pm} = \mu_a + \mu_b \pm \sqrt{(\mu_a - \mu_b)^2 + 2t \cos(k_y)(\mu_a - \mu_b) + 4t^2 \cos^2(k_y) + 4t^2 \cos^2(k_y - k_x/2) + 4t^2 \sin^2(k_x/2)} \quad (63)$$

Apparently, half filling is achieved globally, iff $\mu_a = -\mu_b \equiv \mu$. (The spectrum is symmetric around $\varepsilon = 0$ in this case.) In order to fulfill the local constraint one still has to calculate the local expectation values. It would be weird however, if the chemical potential differed between a and b sites (-i charge density waves?).

$$\Rightarrow \varepsilon(\vec{k})_{\pm} = \pm 2|t| \sqrt{(\mu/(2t) + \cos(k_y))^2 + \cos^2(k_y - k_x/2) + \sin^2(k_x/2)} \quad (64)$$

It is correct that ε only depends on the absolute value of t , it looks like this would completely neutralise the particle-hole transformation. The periodicity is a bit weird here, since the expression is not periodic in 2π in every direction as one would expect. I am pretty sure this comes from the fact that the distance between NN is the same as for the 0-flux case but the Brillouin zone got cut in half in one direction.

represent hamiltonian as

$$H = \sum_{q,\sigma} (a_{q\sigma}^\dagger \quad b_{q\sigma}^\dagger) \vec{h} \cdot \vec{\sigma} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \end{pmatrix} \quad (65)$$

$$\vec{h} = \begin{pmatrix} 2t \cos(k_y - k_x/2) \\ 2t \sin(k_x/2) \\ 2t \cos(k_y) + \mu \end{pmatrix} \quad (66)$$

Then $\varepsilon(\vec{q})_\pm = \pm |\vec{h}|$ and

$$\frac{1}{\sqrt{2\varepsilon_+(\varepsilon_+ + 2t \cos(k_y) + \mu)}} \begin{pmatrix} \varepsilon_+ + 2t \cos(k_y) + \mu & -2t \cos(k_y - k_x/2) + 2ti \sin(k_x/2) \\ 2t \cos(k_y - k_x/2) + 2ti \sin(k_x/2) & \varepsilon_+ + 2t \cos(k_y) + \mu \end{pmatrix} \quad (67)$$

is the transformation matrix U . It has the properties that it is unitary and has determinant one. The expectation values are more involved bc of said matrix transformation. For example

$$\begin{aligned} \langle a_{i\uparrow}^\dagger b_{j\uparrow} \rangle &\stackrel{\text{FT}}{=} \frac{2}{N} \sum_{q,p} e^{i(\vec{q}\vec{R}_j - \vec{p}\vec{R}_i)} \underbrace{\langle a_p^\dagger b_q \rangle}_{\propto \delta_{q,p}} \\ &= \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} \langle a_q^\dagger b_q \rangle \\ &\stackrel{\text{BT}}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} (u_{11}^* u_{21} f(\varepsilon_+) + u_{12}^* u_{22} f(\varepsilon_-)) \end{aligned} \quad (68)$$

Simplify this using

1. $\varepsilon_+ = -\varepsilon_- \Rightarrow f(\varepsilon_+) + f(\varepsilon_-) = 1$
2. $u_{11} \in \mathbb{R}$
3. $u_{11} = u_{22}$
4. $u_{12}^* = -u_{21}$
5. $|u_{11}|^2 + |u_{12}|^2 = 1$ (normalization of eigenvectors)

$$\begin{aligned} u_{11}^* u_{21} f(\varepsilon_+) + u_{12}^* u_{22} f(\varepsilon_-) &\stackrel{1.}{=} (u_{11}^* u_{21} - u_{22} u_{12}^*) f(\varepsilon_+) + u_{22} u_{12}^* \\ &\stackrel{2.,3.,4.}{=} 2u_{11} u_{21} f(\varepsilon_+) - u_{11} u_{21} \\ &= u_{11} u_{21} (2f(\varepsilon_+) - 1) \end{aligned} \quad (69)$$

Due to the unit cell doubling the thermodynamic limit has to be slightly modified $\sum_k \rightarrow ((\Omega N)/2)/(2\pi)^2 \int dk$. (Each unit cell has 2 sites, this is why the factor 1/2 appears.)

$$\begin{aligned} \Rightarrow \langle a_{i\uparrow}^\dagger b_{j\uparrow} \rangle &= \Omega \int_{\text{1.BZ}} \frac{d^2 q}{(2\pi)^2} e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} u_{11} u_{21} (2f(\varepsilon_+) - 1) \\ &= \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \exp \left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] u_{11} u_{21} (2f(\varepsilon_+) - 1) \end{aligned} \quad (70)$$

$$u_{11} u_{21} = (t/\varepsilon)(\cos(k_y - k_x/2) + i \sin(k_x/2)) \quad (71)$$

Now for the other expectation values:

$$\Rightarrow \langle b_{i\uparrow}^\dagger a_{j\uparrow} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \exp \left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] u_{11} u_{21}^* (2f(\varepsilon_+) - 1) \quad (72)$$

$$\begin{aligned} \langle a_{i\uparrow}^\dagger a_{j\uparrow} \rangle &= \dots = \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} [|u_{11}|^2 f(\varepsilon_+) + |u_{12}|^2 f(\varepsilon_-)] \\ &\stackrel{1.}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} [|u_{12}|^2 - |u_{11}|^2] f(\varepsilon_-) + |u_{11}|^2 \\ &\stackrel{5.}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} [(1 - 2u_{11}^2) f(\varepsilon_-) + u_{11}^2] \\ &\stackrel{1.}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} [(1 - 2u_{11}^2)(1 - f(\varepsilon_+)) + u_{11}^2] \end{aligned} \quad (73)$$

$$\Rightarrow \langle a_{i\uparrow}^\dagger a_{j\uparrow} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \exp \left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] [(1 - 2u_{11}^2)(1 - f(\varepsilon_+)) + u_{11}^2] \quad (74)$$

$$\Rightarrow \langle b_{i\uparrow}^\dagger b_{j\uparrow} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \exp \left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] [(1 - 2u_{11}^2) f(\varepsilon_+) + u_{11}^2] \quad (75)$$

$$u_{11}^2 = (1/2\varepsilon_+)(\varepsilon_+ + 2t \cos(k_y) + \mu) \quad (76)$$

$$\langle N^a \rangle = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} (1 - 2u_{11}^2) f(\varepsilon_-) + u_{11}^2 \quad (77)$$

$$\langle N^b \rangle = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} (1 - 2u_{11}^2) f(\varepsilon_+) + u_{11}^2 \quad (78)$$

The constraint $\langle N_i^a \rangle = 1 = \langle N_i^b \rangle$ **should**, as gloal halfiling is already ensured and $\langle N^a \rangle$ and $\langle N^b \rangle$ are uniform by construction, be equivalent to $\langle N^a \rangle = \langle N^b \rangle$. Using the expressions above:

$$\langle N^a \rangle - \langle N^b \rangle = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} (1 - 2u_{11}^2)(1 - 2f(\varepsilon_+)) \stackrel{!}{=} 0 \quad (79)$$

Numerics say that $\langle N^a \rangle = \langle N^b \rangle$ for $\mu = 0$ (and therefore $\mu_a = \mu_b = 0$). **I feel like there is a more sophisticated argument to be made here, mabye even without the use of mathematica. Maybe one can find some symmetry argument using the form of $1 - 2f(\varepsilon_+)$.**

Sanity check:

$$\langle N^a + N^b \rangle = 2 \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} 1 = 2 \quad (80)$$

Calculating the mean field parameters is less straight forward, as the formulae also depend on the types of sites they connect. The formulae are

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} a_{i+\delta_1\sigma} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} e^{ik_y} [(1 - 2u_{11}^2)(1 - f(\varepsilon_+)) + u_{11}^2] \quad (81)$$

$$\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^{\dagger} b_{i+\delta_1\sigma} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} e^{ik_y} [(1 - 2u_{11}^2)f(\varepsilon_+) + u_{11}^2] \quad (82)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_3\sigma} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} e^{i(k_y - k_x/2)} u_{11} u_{21} (2f(\varepsilon_+) - 1) \quad (83)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_2\sigma} \rangle = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} e^{ik_x/2} u_{11} u_{21}^* (2f(\varepsilon_+) - 1) \quad (84)$$

Consistency check: wehn reversing the direction of the bonds the $\vec{\delta}_2$ bond paramter should obtain a minus sign but the $\vec{\delta}_3$ bond parameter not. **I wasnt able to figure this out by just looking at the formulae, so Ill just have to check by looking at the numerics for now.**

Another observation:

$$\begin{aligned} \frac{1}{2} \sum_{\sigma} \left(\langle a_{i\sigma}^{\dagger} a_{i+\delta_1\sigma} \rangle + \langle b_{i\sigma}^{\dagger} b_{i+\delta_1\sigma} \rangle \right) &= \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} e^{ik_y} \\ &= \frac{1}{2\pi} \int_{[-\pi, \pi]} dk_y e^{ik_y} \\ &= \frac{\sin(\pi)}{\pi} \\ &= 0 \end{aligned} \quad (85)$$

$$\Rightarrow \frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} a_{i+\delta_1\sigma} \rangle = -\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^{\dagger} b_{i+\delta_1\sigma} \rangle, \quad (86)$$

as expected. The numerical results are

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} a_{i+\delta_1\sigma} \rangle = 0.19754 \quad (87)$$

$$\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^{\dagger} b_{i+\delta_1\sigma} \rangle = -0.19754 \quad (88)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_3\sigma} \rangle = 0.19754 \quad (89)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_2\sigma} \rangle = 0.19754 \quad (90)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i-\delta_2\sigma} \rangle = -0.19754 \quad (91)$$

The corresponding hopping amplitudes are

$$t(\pm 0.19754) = \mp 0.87690J \quad (92)$$

$$(93)$$

At last I will derive the energy density (since I had trouble with this before).

$$\begin{aligned}
\frac{\langle H \rangle}{N} &= \frac{1}{N} \left\langle \sum_{q\sigma} \varepsilon_{+\sigma} \alpha_{+\sigma}^\dagger \alpha_{+\sigma} + \varepsilon_{-\sigma} \alpha_{-\sigma}^\dagger \alpha_{-\sigma} \right\rangle \\
&\xrightarrow{\text{th. lim.}} \frac{1}{N} \frac{\Omega N}{2} \int_{\text{1.BZ}} \frac{d^2 q}{(2\pi)^2} \varepsilon_+ f(\varepsilon_+) + \varepsilon_- f(\varepsilon_-) \\
&= \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \varepsilon_+ (2f(\varepsilon_+) - 1)
\end{aligned} \tag{94}$$

(the blue 2 in the second line is the spin degeneracy)

Observation: the energy density is invariant under particle-hole transformation

$$\varepsilon_+ (2f(\varepsilon_+) - 1) \xrightarrow{\text{ph-trafo}} \varepsilon_- (2f(\varepsilon_-) - 1) = -\varepsilon_+ (1 - 2f(\varepsilon_+)) = \varepsilon_+ (2f(\varepsilon_+) - 1) \tag{95}$$

The full energy density is:

$$\frac{\langle H \rangle}{N} = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \varepsilon_+(\vec{k}) (2f(\varepsilon_+(\vec{k})) - 1) + 6 \left(\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} \rangle|^2 \right) |\langle \chi_{i,i+\delta} \rangle|^2 \right) = -1.2206 J \tag{96}$$

0.3.2 Pedro's method

Here I apply Pedros method using the terms from my Hamiltonian. The free energy is

$$f_0 \equiv \frac{F_0(\kappa, \chi, T=0)}{N} = \frac{E_0}{N} + \frac{1}{N} \sum_{\vec{k}} \varepsilon(\vec{k}) \tag{97}$$

$$E_0 = 6N \left(\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{ij} \rangle|^2 \right) |\langle \chi_{ij} \rangle|^2 \right) \tag{98}$$

$$\varepsilon(\vec{k}) = -2 \left| -\tilde{J} \left(1 - 4\tilde{\kappa} |\langle \chi_{ij} \rangle|^2 \right) \langle \chi_{ij} \rangle \right| \sqrt{\cos^2(k_1) + \cos^2(k_3) + \sin^2(k_2)} \tag{99}$$

From now on I use the shorthand $|\langle \chi_{ij} \rangle| \equiv \chi$.

$$\begin{aligned}
\Rightarrow f_0 &= \tilde{J} (6(1 - 6\tilde{\kappa}\chi^2)\chi^2 - 2\gamma |\chi(1 - 4\tilde{\kappa}\chi^2)|) \\
&= \tilde{J} (\mp 2\gamma\chi + 6\chi^2 \pm 8\gamma\tilde{\kappa}\chi^3 - 36\tilde{\kappa}\chi^4)
\end{aligned} \tag{100}$$

with

$$\gamma = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \sqrt{\cos^2(k_y) + \cos^2(k_y - k_x/2) + \sin^2(k_x/2)} \approx 1.18524 \tag{101}$$

Compared to Pedros results there is also a factor 2, besides other problems, because Pedro calculates $f/M (f/2)!$

$$\frac{\partial f_0}{\partial \chi} = \tilde{J} (\pm 2\gamma - 12\chi) (12\tilde{\kappa}\chi^2 - 1) \quad \frac{\partial^2 f_0}{\partial \chi^2} = \tilde{J} (12 \pm 48\gamma\tilde{\kappa}\chi - 432\tilde{\kappa}\chi^2) \tag{102}$$

The extremal points of the free energy are at

$$\chi' = \gamma/6, \quad \chi'' = \pm 1/\sqrt{12\tilde{\kappa}} \tag{103}$$

$-1/\sqrt{12\tilde{\kappa}}$ is always a Maximum. $\gamma/6$ is a minimum for $\kappa \leq 6/\gamma^2$, afterwards $1/\sqrt{12\tilde{\kappa}}$ becomes a minimum:

$$f_0 = \begin{cases} \frac{\tilde{J}\gamma^2}{6} \left(\frac{\tilde{\kappa}\gamma^2}{18} - 1 \right), & \tilde{\kappa} \leq 6/\gamma^2 \\ \tilde{J} \left(\frac{1}{4\tilde{\kappa}} - \frac{2\gamma}{9} \sqrt{\frac{3}{\tilde{\kappa}}} \right), & \tilde{\kappa} > 6/\gamma^2 \end{cases} \quad (104)$$

$6/\gamma^2 \approx 2.1355$. For $\kappa = 10 \Rightarrow \tilde{\kappa} = 1.6$ and therefore $\tilde{\kappa} < 6/\gamma^2$. Hence, the free energy minimum is at

$$\chi = \gamma/6 = 0.19754, \quad (105)$$

with the free energy

$$f_0 = -1.22206 J. \quad (106)$$

These correspond to the values I calculated using the pedestrian method above. Mind that under particle-hole transformation the calculation doesn't change (the reason basically being the absolute value in the energy, or more physically the particle-hole symmetry of the system).

0.3.3 fermi surface

The dispersion relation reads

$$\varepsilon(\vec{q}) = 2|t| \sqrt{\cos^2(q_x) + \sin^2 \left(\frac{1}{2} (q_x + \sqrt{3}q_y) \right) + \cos^2 \left(\frac{1}{2} (q_x - \sqrt{3}q_y) \right)} \quad (107)$$

I will start by calculating the fermi surface $\varepsilon(\vec{q}) = 0$, which means that the expression under the square root must vanish. Using

$$\sin^2 \left(\frac{1}{2} (q_x + \sqrt{3}q_y) \right) + \cos^2 \left(\frac{1}{2} (q_x - \sqrt{3}q_y) \right) = 1 + \sin(q_x) \sin(\sqrt{3}q_y) \quad (108)$$

I obtain the following equation for the fermi surface:

$$\underbrace{1 + \cos^2(q_0^x)}_{\in[1,2]} = - \underbrace{\sin(q_0^x) \sin(\sqrt{3}q_0^y)}_{\in[-1,1]}. \quad (109)$$

The equation can be simplified by comparing the ranges of both sides (see above)

$$\Rightarrow 1 + \cos^2(q_0^x) \stackrel{!}{=} 1 \Rightarrow q_0^x = \frac{\pi}{2} + \pi k, \quad k \in \mathbb{Z} \quad (110)$$

$$\Rightarrow \sin(\sqrt{3}q_0^y) = \begin{cases} -1 & k \text{ even} \\ 1 & k \text{ odd} \end{cases} \quad (111)$$

The solution (and thereby the Fermi surface) is

$$q_0^x = \frac{\pi}{2} + \pi k \quad (112)$$

$$q_0^y = \frac{1}{\sqrt{3}} \left(\frac{(-1)^{k+1}\pi}{2} + 2\pi m \right), \quad (113)$$

where $k, m \in \mathbb{Z}$. This means that the fermi surface is a collection of isolated points. More scientifically spoken: the state is a semi-metal. Two of those Fermi-points lie in the first Brillouin zone:

$$\vec{K} = \frac{\pi}{2} \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix} \quad (k = m = 0) \quad (114)$$

$$\vec{K}' = \frac{\pi}{2} \begin{pmatrix} -1 \\ 1/\sqrt{3} \end{pmatrix} \quad (k = -1, m = 0). \quad (115)$$

Mind you that these points are not equivalent! Evidently they are time-reversal partners.

0.3.4 small momentum expansion

To access the low-energy physics we must approximate the Hamiltonian around the fermi-points using

$$f(\vec{q}_0 + \delta\vec{q}) \approx f(\vec{q}_0) + (\nabla_{\vec{q}} f(\vec{q})|_{\vec{q}=\vec{q}_0}) \cdot \delta\vec{q}. \quad (116)$$

Around $\vec{K}$:

$$\cos(\vec{\delta}_1(\vec{K} + \vec{p})) \approx \cos(\vec{\delta}_1\vec{K}) - \sin(\vec{\delta}_1\vec{K})\vec{\delta}_1\vec{p} = -\vec{\delta}_1\vec{p} \quad (117)$$

$$\sin(\vec{\delta}_2(\vec{K} + \vec{p})) \approx \sin(\vec{\delta}_2\vec{K}) + \cos(\vec{\delta}_2\vec{K})\vec{\delta}_2\vec{p} = \vec{\delta}_2\vec{p} \quad (118)$$

$$\cos(\vec{\delta}_3(\vec{K} + \vec{p})) \approx \cos(\vec{\delta}_3\vec{K}) - \sin(\vec{\delta}_3\vec{K})\vec{\delta}_3\vec{p} = -\vec{\delta}_3\vec{p} \quad (119)$$

Use

$$\vec{\delta}_1\vec{K} = \frac{\pi}{2} \quad (120)$$

$$\vec{\delta}_2\vec{K} = 0 \quad (121)$$

$$\vec{\delta}_3\vec{K} = \frac{\pi}{2} \quad (122)$$

to obtain

$$\Rightarrow H_{\vec{K}} \approx 2t \sum_{p,\sigma} \begin{pmatrix} a_{p\sigma}^\dagger & b_{p\sigma}^\dagger \end{pmatrix} \begin{pmatrix} -\vec{\delta}_1\vec{p} & (-\vec{\delta}_3 - i\vec{\delta}_2)\vec{p} \\ (-\vec{\delta}_3 + i\vec{\delta}_2)\vec{p} & \vec{\delta}_1\vec{p} \end{pmatrix} \begin{pmatrix} a_{p\sigma} \\ b_{p\sigma} \end{pmatrix} \quad (123)$$

And around $\vec{K}' = -\vec{K}$:

$$\Rightarrow H_{\vec{K}'} \approx 2t \sum_{p,\sigma} \begin{pmatrix} a_{p\sigma}^\dagger & b_{p\sigma}^\dagger \end{pmatrix} \begin{pmatrix} \vec{\delta}_1\vec{p} & (\vec{\delta}_3 - i\vec{\delta}_2)\vec{p} \\ (\vec{\delta}_3 + i\vec{\delta}_2)\vec{p} & -\vec{\delta}_1\vec{p} \end{pmatrix} \begin{pmatrix} a_{p\sigma} \\ b_{p\sigma} \end{pmatrix} \quad (124)$$

This gives the dispersion relation (for both cones)

$$\begin{aligned} \varepsilon &\approx 2t \sqrt{p_x^2 + \frac{1}{4} \left(p_x + \sqrt{3}p_y \right)^2 + \frac{1}{4} \left(p_x - \sqrt{3}p_y \right)^2} \\ &= \sqrt{6}t \sqrt{p_x^2 + p_y^2} \\ &= \sqrt{6}t |\vec{p}| \end{aligned} \quad (125)$$

Thus, the fermi velocity is $v_F = \sqrt{6}t$. This result has been reported before here [5]. $\vec{K}$ and $\vec{K}'$ are time reversel partners, thus the two hamiltonains should map onto each other under time revearesal. I would guess that $T = -C$ in k-space, where C is complex conjugation, which would be consistent with the form of the Hamiltonian.

0.3.5 Continuum Model/ Gradient Expansion

Next I will derive the low-energy Hamiltonian in real space, as a continuum theory (/field theory). The real space Hamiltonian reads (copied from above with the chemical potentials set to zero)

$$H = \frac{t}{2} \sum_{i\sigma} \left(a_{i\sigma}^\dagger a_{i+\delta_1;\sigma} + a_{i\sigma}^\dagger a_{i-\delta_1;\sigma} - a_{i\sigma}^\dagger b_{i+\delta_2;\sigma} + a_{i\sigma}^\dagger b_{i-\delta_2;\sigma} + a_{i\sigma}^\dagger b_{i+\delta_3;\sigma} + a_{i\sigma}^\dagger b_{i-\delta_3;\sigma} \right. \\ \left. - b_{i\sigma}^\dagger b_{i+\delta_1;\sigma} - b_{i\sigma}^\dagger b_{i-\delta_1;\sigma} + b_{i\sigma}^\dagger a_{i+\delta_2;\sigma} - b_{i\sigma}^\dagger a_{i-\delta_2;\sigma} + b_{i\sigma}^\dagger a_{i+\delta_3;\sigma} + b_{i\sigma}^\dagger a_{i-\delta_3;\sigma} \right) \quad (126)$$

The idea is to decompose the the real-space wavefunction by its momentum - one part is the momentum is the momentum of the dirac point, the other point describes a small deviation from the dirac point (small wavevector - i large wavelength - i slowly varying in real space). This justifies a gradient expansion of the "deviation field". **think of a better explanation for this. But I guess its correct in its spirit.**

$$a_{i\sigma} \sim e^{i\vec{K}\vec{R}_i} A_{K\sigma}(\vec{R}_i) + e^{i\vec{K}'\vec{R}_i} A_{K'\sigma}(\vec{R}_i) \\ b_i \sim e^{i\vec{K}\vec{R}_i} B_{K\sigma}(\vec{R}_i) + e^{i\vec{K}'\vec{R}_i} B_{K'\sigma}(\vec{R}_i) \quad (127)$$

$A_{K\sigma}, A_{K'\sigma}, B_{K\sigma}$ and $B_{K'\sigma}$ are slowly varying fermionic fields. I begin with the expansion around the K -point. **If the above expansion is correct, the cross terms should vanish. Why?** The prototypical method is

$$a_{i\sigma}^\dagger a_{i+\delta;\sigma} \sim e^{i\vec{K}\vec{\delta}} A_{K\sigma}^\dagger(\vec{R}_i) A_{K\sigma}(\vec{R}_i + \vec{\delta}) \\ \approx e^{i\vec{K}\vec{\delta}} A_{K\sigma}^\dagger(\vec{R}_i) \left(1 + \vec{\delta} \vec{\nabla}_{\vec{R}} \right) A_{K\sigma}(\vec{R}_i) \\ = e^{i\vec{K}\vec{\delta}} A_{K\sigma}^\dagger(\vec{R}_i) \left(1 + i\vec{\delta} \vec{p} \right) A_{K\sigma}(\vec{R}_i), \quad (128)$$

where $\vec{p} = -i\vec{\nabla}_{\vec{R}}$. The diagonal terms are

$$A_{K\sigma}^\dagger(\vec{R}_i) \left[e^{i\vec{K}\vec{\delta}_1} (1 + i\vec{\delta}_1 \vec{p}) + e^{-i\vec{K}\vec{\delta}_1} (1 - i\vec{\delta}_1 \vec{p}) \right] A_{K\sigma}(\vec{R}_i) \quad (129)$$

$$B_{K\sigma}^\dagger(\vec{R}_i) \left[-e^{i\vec{K}\vec{\delta}_1} (1 + i\vec{\delta}_1 \vec{p}) - e^{-i\vec{K}\vec{\delta}_1} (1 - i\vec{\delta}_1 \vec{p}) \right] B_{K\sigma}(\vec{R}_i). \quad (130)$$

The off-diagonal terms are

$$A_{K\sigma}^\dagger(\vec{R}_i) \frac{t}{2} \left[-e^{i\vec{\delta}_2 \vec{K}} (1 + i\vec{\delta}_2 \vec{p}) + e^{-i\vec{\delta}_2 \vec{K}} (1 - i\vec{\delta}_2 \vec{p}) + e^{i\vec{\delta}_3 \vec{K}} (1 + i\vec{\delta}_3 \vec{p}) + e^{-i\vec{\delta}_3 \vec{K}} (1 - i\vec{\delta}_3 \vec{p}) \right] B_{K\sigma}(\vec{R}_i) \quad (131)$$

$$B_{K\sigma}^\dagger(\vec{R}_i) \frac{t}{2} \left[e^{i\vec{\delta}_2 \vec{K}} (1 + i\vec{\delta}_2 \vec{p}) - e^{-i\vec{\delta}_2 \vec{K}} (1 - i\vec{\delta}_2 \vec{p}) + e^{i\vec{\delta}_3 \vec{K}} (1 + i\vec{\delta}_3 \vec{p}) + e^{-i\vec{\delta}_3 \vec{K}} (1 - i\vec{\delta}_3 \vec{p}) \right] A_{K\sigma}(\vec{R}_i) \quad (132)$$

Use

$$\vec{\delta}_1 \vec{K} = \frac{\pi}{2} \quad (133)$$

$$\vec{\delta}_2 \vec{K} = 0 \quad (134)$$

$$\vec{\delta}_3 \vec{K} = \frac{\pi}{2} \quad (135)$$

to obtain

$$H_{\vec{K}} \approx \sum_{\sigma} 2t \int d^2 \vec{R} \quad \Psi_{K\sigma}^{\dagger}(\vec{R}) \begin{pmatrix} -\vec{\delta}_1 \vec{p} & (-\vec{\delta}_3 - i\vec{\delta}_2) \vec{p} \\ (-\vec{\delta}_3 + i\vec{\delta}_2) \vec{p} & \vec{\delta}_1 \vec{p} \end{pmatrix} \Psi_{K\sigma}(\vec{R}), \quad (136)$$

with

$$\Psi_{K\sigma}(\vec{R}) = \begin{pmatrix} A_{K\sigma}(\vec{R}) \\ B_{K\sigma}(\vec{R}) \end{pmatrix}. \quad (137)$$

Maybe find a few things to write about Isospin? However this is probably not the real Isospin of theory yet as H is not diagonal.

Doing the same around $\vec{K}'$ gives

$$H_{\vec{K}'} \approx \sum_{\sigma} 2t \int d^2 \vec{R} \quad \Psi_{K'\sigma}^{\dagger}(\vec{R}) \begin{pmatrix} \vec{\delta}_1 \vec{p} & (\vec{\delta}_3 - i\vec{\delta}_2) \vec{p} \\ (\vec{\delta}_3 + i\vec{\delta}_2) \vec{p} & -\vec{\delta}_1 \vec{p} \end{pmatrix} \Psi_{K'\sigma}(\vec{R}) \quad (138)$$

Maybe it is handy to write $\vec{\delta}_1 = \vec{\delta}_2 + \vec{\delta}_3$. When Comparing this to the results obtained by expanding in reciprocal space, keep in mind that here $\vec{p}$ is an operator. So technically, one would have to do a kind of fourier transform for the fermionic fields to find the exact representation as found in reciprocal space. This is why the Hamiltonian looks so similar, even though we are still in real space here.

Imagine we did this **small frequency**? Fourier expansion and we found the reciprocal space Hamiltonian (replace the momentum operator by its eigenvalue!), the corresponding disperion relation would then read

$$\varepsilon(\vec{p}) = \left| 2t \begin{pmatrix} \mp \vec{\delta}_3 \vec{p} \\ \vec{\delta}_2 \vec{p} \\ \mp \vec{\delta}_3 \vec{p} \end{pmatrix} \right| = \sqrt{6}t \sqrt{p_x^2 + p_y^2} = \sqrt{6}t |\vec{p}|, \quad (139)$$

which complies with the k-space result. In case I want to diagonalize this in operator form at some point; since the problem is non-interacting there must be a harmonic oscillator hidden in there.

0.4 0- π State on Strained Lattice

references are [4], [3], [6], [1] and [7]

0.4.1 Basics of Strain

For the purpose of introducing strain into the model, I first establish its effect on the coordinates and exchange couplings **Awkward formulation**

$$\vec{R}_i \mapsto \vec{R}'_i = \vec{R}_i + \vec{U}(\vec{R}_i) \quad (140)$$

$$\vec{\delta}_{ij} \mapsto \vec{\delta}'_{ij} = \vec{\delta}_{ij} + \vec{U}(\vec{R}_j) - \vec{U}(\vec{R}_i) \quad (141)$$

$$\vec{\delta}_{ij} = \vec{R}_j - \vec{R}_i ;)$$

$\vec{U}(\vec{R})$ is the displacement field. We approximate that $\vec{U}$ varies slowly with $\vec{R}$:

$$\vec{U}(\vec{R}_j) = \vec{U}(\vec{R}_i + \vec{\delta}_{ij}) \approx \vec{U}(\vec{R}_i) + \left(\vec{\delta}_{ij} \vec{\nabla}_{\vec{R}} \right) \vec{U}(\vec{R})|_{\vec{R}_i} \quad (142)$$

$$(143)$$

It is convenient to redefine

$$\left(\vec{\delta}_{ij} \vec{\nabla}_{\vec{R}}\right) \vec{U}(\vec{R})|_{\vec{R}_i} = \underline{u} \cdot \vec{\delta}_{ij}, \quad (144)$$

where $u_{\alpha\beta} = \partial_\beta U_\alpha$ is the strain tensor.

$$\Rightarrow \vec{\delta}'_{ij} \approx \vec{\delta}_{ij} + \underline{u} \cdot \vec{\delta}_{ij} \quad (145)$$

I do the same approximation for the exchange couplings

$$J(\vec{\delta}_{ij}) \mapsto J(\vec{\delta}'_{ij}) \approx J\left(\vec{\delta}_{ij} + \underline{u} \cdot \vec{\delta}_{ij}\right) \quad (146)$$

$$\approx J(\vec{\delta}_{ij}) + \left(\underline{u} \cdot \vec{\delta}_{ij}\right) \cdot \left(\vec{\nabla}_{\vec{\delta}} J(\vec{\delta})|_{\vec{\delta}_{ij}}\right) \quad (147)$$

For the function $J(\vec{\delta})$ one usually uses the form

$$J(\vec{\delta}) = J \exp\left[-\beta\left(|\vec{\delta}|/a - 1\right)\right] \approx J\left(1 - \beta\left(|\vec{\delta}|/a - 1\right)\right), \quad (148)$$

where a is the lattice constant and $\beta = -d \ln(J)/d \ln(|\vec{\delta}|)|_a$ the magneto-elastic coupling. This form is essentially motivated by the large U-limit of the Hubbard model. In this limit the exchange coupling is proportional to the overlap of the orbitals of neighboring atoms. In this sense one can approximate those as two exponential functions, that are decreasing away from the position of their respective atoms. Therefore, their overlap also decreases exponentially.

$$\begin{aligned} &\Rightarrow \vec{\nabla}_{\vec{\delta}} J(\vec{\delta})|_{\vec{\delta}_{ij}} \stackrel{|\vec{\delta}_{ij}|=a}{=} -\frac{J\beta}{a^2} \vec{\delta}_{ij} \\ &\Rightarrow J(\vec{\delta}'_{ij}) \approx J(\vec{\delta}_{ij}) - \frac{J\beta}{a^2} \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij}\right) \end{aligned} \quad (149)$$

Looking at the matrix elements (129 and below), one also requires the expansion of $\exp\left[i\vec{\delta}_{ij} \vec{K}\right]$. To summarize:

$$\vec{\delta}'_{ij} \approx \vec{\delta}_{ij} + \underline{u} \cdot \vec{\delta}_{ij} \quad (150)$$

$$J(\vec{\delta}'_{ij}) \approx J - \frac{J\beta}{a^2} \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij}\right) \quad (151)$$

$$e^{i\vec{K} \cdot \vec{\delta}'_{ij}} \approx e^{i\vec{K} \cdot \vec{\delta}_{ij}} + ie^{i\vec{K} \cdot \vec{\delta}_{ij}} \left(\vec{K} \cdot \underline{u} \cdot \vec{\delta}_{ij}\right) \quad (152)$$

In this approximation the Mean field does not react to the strain at all??

0.4.2 Straining the 0- π -State

The prototypical form of the matrix elements is (expanding around $\vec{K}$)

$$J(\vec{\delta}_{ij}) a_{i\sigma}^\dagger a_{i+\delta_{ij};\sigma} \sim A_{K\sigma}^\dagger \left(J(\vec{\delta}_{ij}) e^{i\vec{K} \cdot \vec{\delta}_{ij}} (1 + i\vec{\delta}_{ij} \cdot \vec{p})\right) A_{K\sigma} \quad (153)$$

Expanding up to first order in strain I acquire the following terms

$J e^{i\vec{K} \cdot \vec{\delta}_{ij}} (1 + i\vec{\delta}_{ij} \cdot \vec{p})$	unstrained Hamiltonian/ zeroth order
$i J e^{i\vec{K} \cdot \vec{\delta}_{ij}} (\vec{K} \cdot \underline{u} \cdot \vec{\delta}_{ij})$	contribution to $\vec{A}$ of $\mathcal{O}(\underline{u}^1)$
$-\frac{J\beta}{a^2} e^{i\vec{K} \cdot \vec{\delta}_{ij}} (\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij})$	contribution to $\vec{A}$ of $\mathcal{O}(\underline{u}^1)$
$-J e^{i\vec{K} \cdot \vec{\delta}_{ij}} (\vec{K} \cdot \underline{u} \cdot \vec{\delta}_{ij}) (\vec{\delta}_{ij} \cdot \vec{p})$	correction to v_F of $\mathcal{O}(\underline{u}^1)$
$-i \frac{J\beta}{a^2} e^{i\vec{K} \cdot \vec{\delta}_{ij}} (\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij}) (\vec{\delta}_{ij} \cdot \vec{p})$	correction to v_F of $\mathcal{O}(\underline{u}^1)$
$i J e^{i\vec{K} \cdot \vec{\delta}_{ij}} (\underline{u} \cdot \vec{\delta}_{ij}) \cdot \vec{p}$	correction to v_F of $\mathcal{O}(\underline{u}^1)$

(I drop the lattice constant again from now on) I will focus on the terms of order zero + terms of either order 1 in field gradient or order 1 in strain [4]. **I havent come up with a good argument for this yet. How does one even compares orders between different expansions?** This gives

$$\begin{aligned}
J(\vec{\delta}_{ij}) e^{i\vec{K} \cdot \vec{\delta}_{ij}} (1 + i\vec{\delta}_{ij} \cdot \vec{p}) &\sim J e^{i\vec{K} \cdot \vec{\delta}_{ij}} \left(1 + i\vec{\delta}_{ij} \cdot \vec{p} + i \left(\vec{K} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) - \beta \left(\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij} \right) \right) \\
&= J e^{i\vec{K} \cdot \vec{\delta}_{ij}} \left(1 + i\vec{\delta}_{ij} \cdot \left[\vec{p} + \underline{u}^T \left(\vec{K} + i\beta \vec{\delta}_{ij} \right) \right] \right)
\end{aligned} \tag{154}$$

Hence, on this level

$$\vec{A} = -\underline{u}^T \left(\vec{K} + i\beta \vec{\delta}_{ij} \right) \tag{155}$$

(the minus comes from minimal coupling). It turns out however, that $\vec{B} = 0$, which is explained in what follows. In general, however

$$B = (\partial_x A_y - \partial_y A_x) \tag{156}$$

The first term is simply rotationless (this was shown here [6]) and can therefore be "gauged away".

Regarding the second term; I take a step back to the dispersion relation of the unstrained model

$$H = \sum_{\vec{q}\sigma} \Psi_{q\sigma}^\dagger \cdot \left(\vec{h}(\vec{q}) \cdot \vec{\sigma} \right) \cdot \Psi_{q\sigma} \quad \text{with} \quad \vec{h}(\vec{q}) = 2t \begin{pmatrix} \sin(\vec{q} \cdot \vec{\delta}_3) \\ \cos(\vec{q} \cdot \vec{\delta}_2) \\ \cos(\vec{q} \cdot \vec{\delta}_1) \end{pmatrix} \Rightarrow \varepsilon(\vec{q}) = |\vec{h}(\vec{q})|. \tag{157}$$

By definition, $\varepsilon(\vec{K}^{(l)}) = 0$. For this to be true, every component of $\vec{h}$ must vanish individually at $\vec{K}^{(l)}$. This gives three conditions for $\vec{q}$, one for each direction.

Evidently each expression in $\vec{h}$ is obtained by adding contributions from hopping along $\vec{\delta}_{ij}$ and along the opposite direction $(-\vec{\delta}_{ij})$ (this can be seen nicely by considering that in the directions of $\vec{\delta}_3$ and $\vec{\delta}_2$ one simply has tight-binding chains, leading to the well-known cosine dispersion).

For the strain expansion one would have to break these terms (e.g. the cosine terms) up again into the contributions from hopping in opposite directions. However, since $\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij}$ is invariant under $\vec{\delta}_{ij} \mapsto -\vec{\delta}_{ij}$ it acquires the same dispersion-like prefactor as the zeroth order, which vanishes by definition at $\vec{K}^{(l)}$.

So why does the term vanish? Its a mix of the special structure of the hamiltonian (i.e. the local directions decouple in the pauli-matrix basis) and the fact that the expansion term $\vec{\delta}_{ij} \cdot \underline{u} \cdot \vec{\delta}_{ij}$ cant

distinguish between $\vec{\delta}_{ij}$ and $-\vec{\delta}_{ij}$. The same should hold for the π -flux square lattice. Graphene shows none of these features (the symmetry of the term doesn't change of course but there is simply no hopping in the opposite direction). For Kagome the first feature is absent.

It is not clear to me how this can be upheld under gauge transformations (wouldn't this mix directions in the Pauli-matrix basis?)

I rename $\vec{R}' \rightarrow \vec{R}$ and get

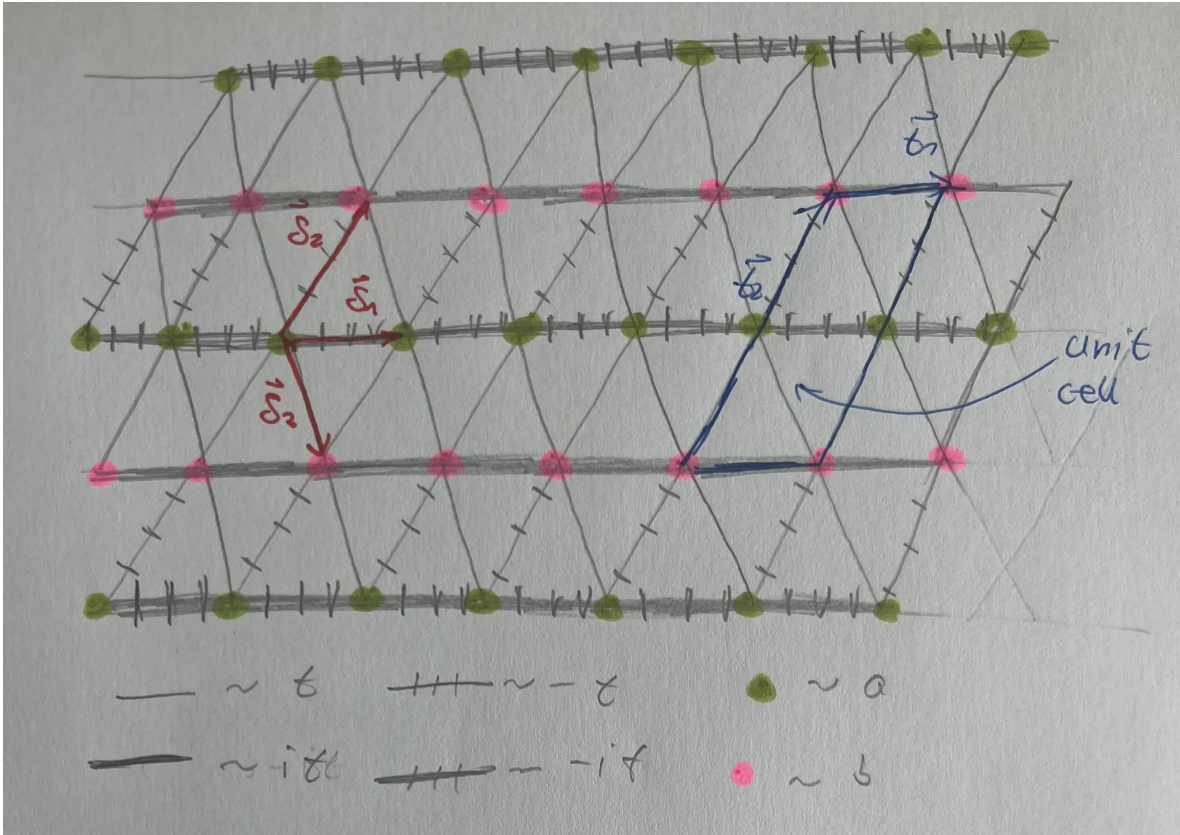
$$H_{\vec{K}} \approx \sum_{\sigma} 2t \int d^2 \vec{R} \quad \Psi_{K\sigma}^{\dagger}(\vec{R}) \begin{pmatrix} -\vec{\delta}_1 (\vec{p} + \underline{u}^T \cdot \vec{K}) & (-\vec{\delta}_3 - i\vec{\delta}_2) (\vec{p} + \underline{u}^T \cdot \vec{K}) \\ (-\vec{\delta}_3 + i\vec{\delta}_2) (\vec{p} + \underline{u}^T \cdot \vec{K}) & \vec{\delta}_1 (\vec{p} + \underline{u}^T \cdot \vec{K}) \end{pmatrix} \Psi_{K\sigma}(\vec{R}), \quad (158)$$

Doing the same procedure around $\vec{K}' = -\vec{K}$ I get

$$H_{\vec{K}'} \approx \sum_{\sigma} 2t \int d^2 \vec{R} \quad \Psi_{K'\sigma}^{\dagger}(\vec{R}) \begin{pmatrix} \vec{\delta}_1 (\vec{p} + \underline{u}^T \cdot \vec{K}') & (\vec{\delta}_3 - i\vec{\delta}_2) (\vec{p} + \underline{u}^T \cdot \vec{K}') \\ (\vec{\delta}_3 + i\vec{\delta}_2) (\vec{p} + \underline{u}^T \cdot \vec{K}') & -\vec{\delta}_1 (\vec{p} + \underline{u}^T \cdot \vec{K}') \end{pmatrix} \Psi_{K'\sigma}(\vec{R}) \quad (159)$$

I would expect to find a pseudo-magnetic field in higher orders of strain?

0.5 $\pi/2$ - $\pi/2$ and $-\pi/2$ - $-\pi/2$ States



0.5.1 Pedestrian Method

This is very similar to the $\pi - 0$ -flux-state. The bare Hamiltonian is

$$H = \sum_{i,\sigma} \left(\frac{t}{2} \left[-ia_{i\sigma}^\dagger a_{i+\delta_1;\sigma} + ia_{i\sigma}^\dagger a_{i-\delta_1;\sigma} - a_{i\sigma}^\dagger b_{i+\delta_2;\sigma} + a_{i\sigma}^\dagger b_{i-\delta_2;\sigma} + a_{i\sigma}^\dagger b_{i+\delta_3;\sigma} + a_{i\sigma}^\dagger b_{i-\delta_3;\sigma} \right. \right. \\ \left. \left. + ib_{i\sigma}^\dagger b_{i+\delta_1;\sigma} - ib_{i\sigma}^\dagger b_{i-\delta_1;\sigma} + b_{i\sigma}^\dagger a_{i+\delta_2;\sigma} - b_{i\sigma}^\dagger a_{i-\delta_2;\sigma} + a_{i\sigma}^\dagger b_{i+\delta_3;\sigma} + a_{i\sigma}^\dagger b_{i-\delta_3;\sigma} \right] \right. \\ \left. + \mu_a a_{i\sigma}^\dagger a_{i\sigma} + \mu_b b_{i\sigma}^\dagger b_{i\sigma} \right) - \frac{N(\mu_a + \mu_b)}{2} \quad (160)$$

The site index notation is not ideal, as a and b sites by definition cant have the same site indices.

$$\Rightarrow H = \sum_{q,\sigma} \begin{pmatrix} a_{q\sigma}^\dagger & b_{q\sigma}^\dagger \end{pmatrix} \begin{pmatrix} 2t \sin(\vec{q}\vec{\delta}_1) + \mu_a & 2t \cos(\vec{q}\vec{\delta}_3) - 2it \sin(\vec{q}\vec{\delta}_2) \\ 2t \cos(\vec{q}\vec{\delta}_3) + 2it \sin(\vec{q}\vec{\delta}_2) & -2t \sin(\vec{q}\vec{\delta}_1) + \mu_b \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \end{pmatrix}. \quad (161)$$

Maybe I should show explicitly that $\mu = 0$, I guess $\mu_a = -\mu_b = \mu$ is clear

Using $\mu_a = \mu_b = 0$:

$$\vec{h} = 2t \begin{pmatrix} \cos(\vec{q}\vec{\delta}_1) \\ \sin(\vec{q}\vec{\delta}_2) \\ \sin(\vec{q}\vec{\delta}_3) \end{pmatrix} \mapsto 2t \begin{pmatrix} \cos(k_y - k_x/2) \\ \sin(k_x/2) \\ \sin(k_y) \end{pmatrix} \quad (162)$$

$$\varepsilon(\vec{k})_\pm = \pm 2|t| \sqrt{\sin^2(k_y) + \cos^2(k_y - k_x/2) + \sin^2(k_x/2)} \quad (163)$$

$$\frac{1}{\sqrt{2\varepsilon_+(\varepsilon_+ + 2t \sin(k_y))}} \begin{pmatrix} \varepsilon_+ + 2t \sin(k_y) & -2t \cos(k_y - k_x/2) + 2ti \sin(k_x/2) \\ 2t \cos(k_y - k_x/2) + 2ti \sin(k_x/2) & \varepsilon_+ + 2t \sin(k_y) \end{pmatrix} \quad (164)$$

$$u_{11}u_{21} = (t/\varepsilon_+)(\cos(k_y - k_x/2) + i \sin(k_x/2)) \quad (165)$$

$$u_{11}^2 = (1/2\varepsilon_+)(\varepsilon_+ + 2t \sin(k_y)) \quad (166)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^\dagger a_{i+\delta_1\sigma} \rangle = \int_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y} [(1 - 2u_{11}^2)(1 - f(\varepsilon_+)) + u_{11}^2] \quad (167)$$

$$\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^\dagger b_{i+\delta_1\sigma} \rangle = \int_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y} [(1 - 2u_{11}^2)f(\varepsilon_+) + u_{11}^2] \quad (168)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^\dagger b_{i+\delta_3\sigma} \rangle = \int_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{i(k_y - k_x/2)} u_{11}u_{21}(2f(\varepsilon_+) - 1) \quad (169)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^\dagger b_{i+\delta_2\sigma} \rangle = \int_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_x/2} u_{11}u_{21}^*(2f(\varepsilon_+) - 1) \quad (170)$$

One has to be a bit careful of where the sign of t enters when choosing $t = 1$ for the self consistency. Due to the complex conjugation the σ_z -terms are insensitive to the sign of t , but the (real) σ_x - and σ_y -terms are not. The results (choosing $t = 1$ are):

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} a_{i+\delta_1\sigma} \rangle = -0.200169i \quad (171)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} a_{i-\delta_1\sigma} \rangle = 0.200169i \quad (172)$$

$$\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^{\dagger} b_{i+\delta_1\sigma} \rangle = 0.200169i \quad (173)$$

$$\frac{1}{2} \sum_{\sigma} \langle b_{i\sigma}^{\dagger} b_{i-\delta_1\sigma} \rangle = -0.200169i \quad (174)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_3\sigma} \rangle = 0.200169 \quad (175)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i-\delta_3\sigma} \rangle = 0.200169 \quad (176)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i+\delta_2\sigma} \rangle = 0.200169 \quad (177)$$

$$\frac{1}{2} \sum_{\sigma} \langle a_{i\sigma}^{\dagger} b_{i-\delta_2\sigma} \rangle = -0.200169 \quad (178)$$

This complies with the cartoon on top. The value for t is

$$|t_{ij}| = \left| J \langle \chi_{ij} \rangle^* (1 + \kappa/2 - 4\kappa |\langle \chi_{ij} \rangle|^2) \right|^{\kappa \equiv 10} \approx 0.88020 J. \quad (179)$$

The free energy is

$$\frac{\langle H \rangle}{N} = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \varepsilon_+(\vec{k}) (2f(\varepsilon_+(\vec{k})) - 1) + 6 \left(\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} \rangle|^2 \right) |\langle \chi_{i,i+\delta} \rangle|^2 \right) = -1.24978 J \quad (180)$$

0.5.2 Pedro's Method

The expression for the free energy is identical to the $0 - \pi$ -state except that γ gets replaced by a different constant α :

$$\alpha = \int_{[-\pi, \pi]^2} \frac{d^2 k}{(2\pi)^2} \sqrt{\sin^2(k_y) + \cos^2(k_y - k_x/2) + \sin^2(k_x/2)} \approx 1.20102 \quad (181)$$

Crucially α is larger than gamma. The results for this method are

$$|\chi| = \alpha/6 = 0.20017, \quad (182)$$

$$f_0 = \tilde{J} \left(\frac{1}{4\tilde{\kappa}} - \frac{2\alpha}{9} \sqrt{\frac{3}{\tilde{\kappa}}} \right) = -1.24979 J \quad (183)$$

The methods agree.

0.5.3 Chern number

Here I will calculate the Chern number associated with the state. The relevant formula for a 2-band system is

$$C = \frac{1}{4\pi} \int_{1.BZ} d^2q \hat{h} \cdot \left(\partial_{q_x} \hat{h} \times \partial_{q_y} \hat{h} \right), \quad (184)$$

where $\hat{h}$ is the normalized Bloch vector (is this the correct terminology?).

$$\vec{h} = 2t \begin{pmatrix} \cos\left(\frac{1}{2}(q_x - \sqrt{3}q_y)\right) \\ \sin\left(\frac{1}{2}(q_x + \sqrt{3}q_y)\right) \\ \sin(q_x) \end{pmatrix} \equiv 2t \begin{pmatrix} \cos(\delta_3) \\ \sin(\delta_2) \\ \sin(\delta_1) \end{pmatrix}, \quad |\vec{h}| \equiv \varepsilon(\vec{q}) \quad (185)$$

I first simplify 184 a bit using

$$\partial_\alpha \hat{h} = \frac{1}{|\vec{h}|} \left(\partial_\alpha \vec{h} - (\hat{h} \cdot \partial_\alpha \vec{h}) \hat{h} \right). \quad (186)$$

When applying this to 184, 3 out of the four terms that appear are proportional to $\hat{h} \times \hat{h} = \vec{0}$.

$$\Rightarrow C = \frac{1}{4\pi} \int_{1.BZ} d^2q [\varepsilon(\vec{q})]^{-3} \vec{h} \cdot \left(\partial_{q_x} \vec{h} \times \partial_{q_y} \vec{h} \right) \quad (187)$$

$$\partial_{q_x} \vec{h} = t \begin{pmatrix} -\sin(\delta_3) \\ \cos(\delta_2) \\ 2\cos(\delta_1) \end{pmatrix}, \quad \partial_{q_y} \vec{h} = \sqrt{3}t \begin{pmatrix} \sin(\delta_1) \\ \cos(\delta_2) \\ 0 \end{pmatrix} \quad (188)$$

$$\begin{aligned} \Rightarrow \vec{h} \cdot \left(\partial_{q_x} \vec{h} \times \partial_{q_y} \vec{h} \right) &= 4\sqrt{3}t^3 (-\cos(\delta_1) \cos(\delta_2) \cos(\delta_3) + \cos(\delta_1) \sin(\delta_2) \sin(\delta_3) - \sin(\delta_1) \cos(\delta_2) \sin(\delta_3)) \\ &= 4\sqrt{3}t^3 (\sin(\delta_1) \sin(\delta_2) \cos(\delta_3) - 1) \end{aligned} \quad (189)$$

$$\Rightarrow C = \frac{\sqrt{3}}{8\pi} \int_{1.BZ} d^2q \frac{\sin(\delta_1) \sin(\delta_2) \cos(\delta_3) - 1}{(\sin^2(\delta_1) + \sin^2(\delta_2) + \cos^2(\delta_3))^{3/2}} \quad (190)$$

For evaluation I use the transformation:

$$\begin{pmatrix} q_x \\ q_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1/\sqrt{3} & -1/\sqrt{3} \end{pmatrix} \begin{pmatrix} k_x \\ k_y \end{pmatrix} \quad (191)$$

$$\Rightarrow C = \frac{1}{8\pi} \int_{[-\pi, \pi]^2} d^2k \frac{\sin(k_y) \sin(k_y - k_x/2) \cos(k_x/2) - 1}{(\sin^2(k_y) + \sin^2(k_y - k_x/2) + \cos^2(k_x/2))^{3/2}} \quad (192)$$

The result is $C = -1$.

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